Conservation Laws

General form:

$$\begin{cases} u_t + [F(u)]_x = 0 \\ u(x,0) = g(x) \end{cases}$$

Usually, F is differentiable, so the PDE is

$$u_t + F'(u)u_x = 0$$

This is a nonlinear transport equation

(Linear transport:

$$u_t + a u_x = 0$$

a = constant or $a(t,x)$ a function.)

↳ slope of char. curves determined by a.

Pick $F(u) = \frac{1}{2}u^2$

$F'(u) = u$

Burger's Equation

$$u_t + u u_x = 0$$

Last time: initial data with a jump

$$g(x) = \begin{cases} u_\ell, & x < 0, \\ u_r, & x \geq 0 \end{cases}$$

We saw:

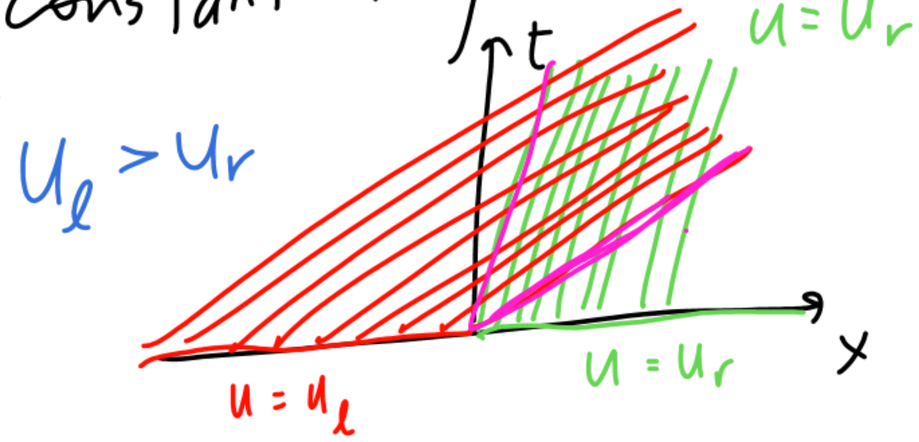
$$z(s) = u(x(s), t(s))$$

$$\frac{dz}{ds} = u_x \frac{dx}{ds} + u_t \frac{dt}{ds} = u_t + u u_x = 0$$

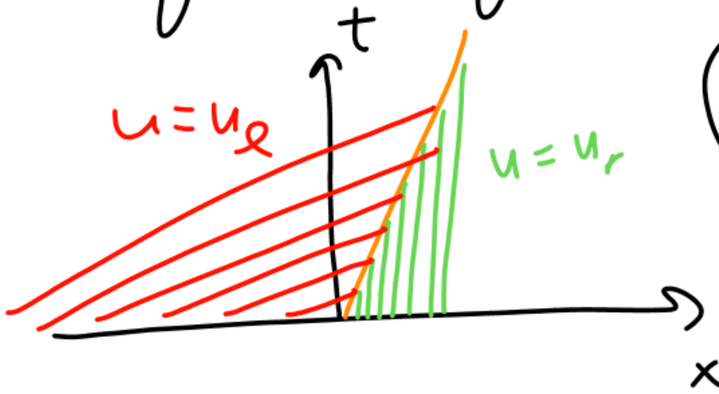
$$\frac{dt}{ds} = 1$$

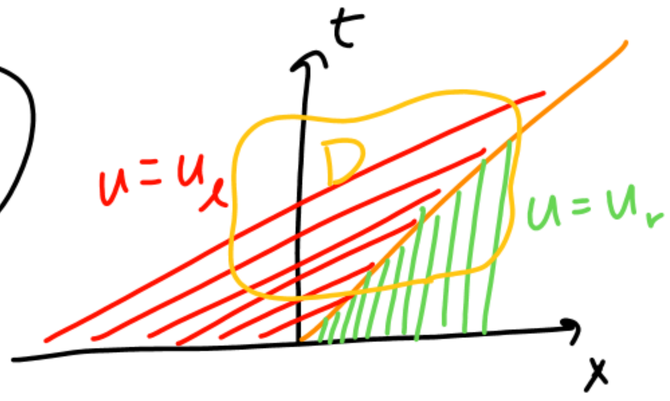
$$\frac{dx}{ds} = u(x(s), t(s))$$

Solution constant along char. curves.



To find where the solution jumps, the PDE doesn't give enough info.



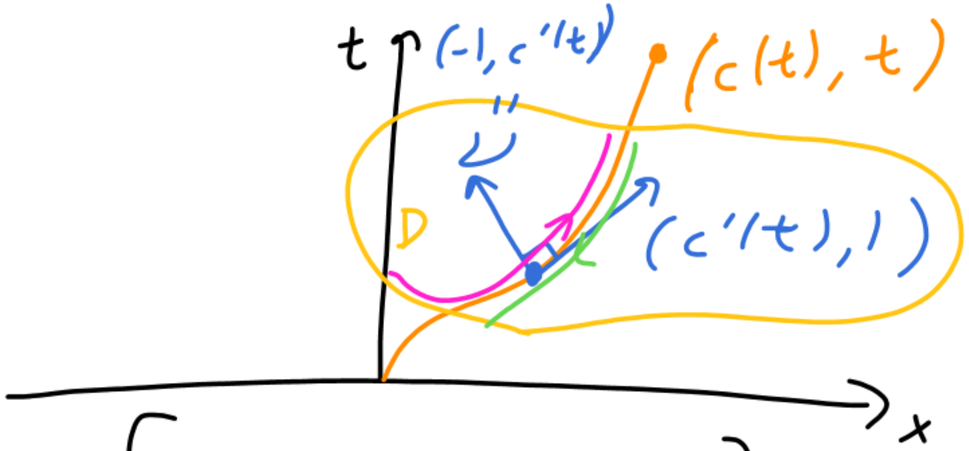


Both are solutions to PDE (outside of jumps)

Weak form of the PDE:

u solves $u_t + [f(u)]_x = 0$ <u>weakly</u> if, for all smooth functions $\phi(x,t)$ with $\phi \equiv 0$ outside a compact subset D of $(0,\infty) \times \mathbb{R}_x$,

$$\int_0^\infty \int_{-\infty}^\infty \left[u\phi_t + F(u)\phi_x \right] dx\,dt = 0$$



$$\int_C \Big[\underbrace{\big[F(u_\ell) - F(u_r)\big]\nu^1 + (u_\ell - u_r)\nu^2}_{=0 \text{ along } C} \Big] \phi \, ds = 0$$

$$\big(F(u_\ell) - F(u_r)\big)(-1) + (u_\ell - u_r)\, c'(t) = 0$$

$$c'(t) = \frac{F(u_\ell) - F(u_r)}{u_\ell - u_r} = \frac{\text{jump in } F}{\text{jump in } u}$$

$$c'(t) = \frac{F(u_\ell) - F(u_r)}{u_\ell - u_r}$$

$$F(u) = \frac{1}{2}u^2,$$

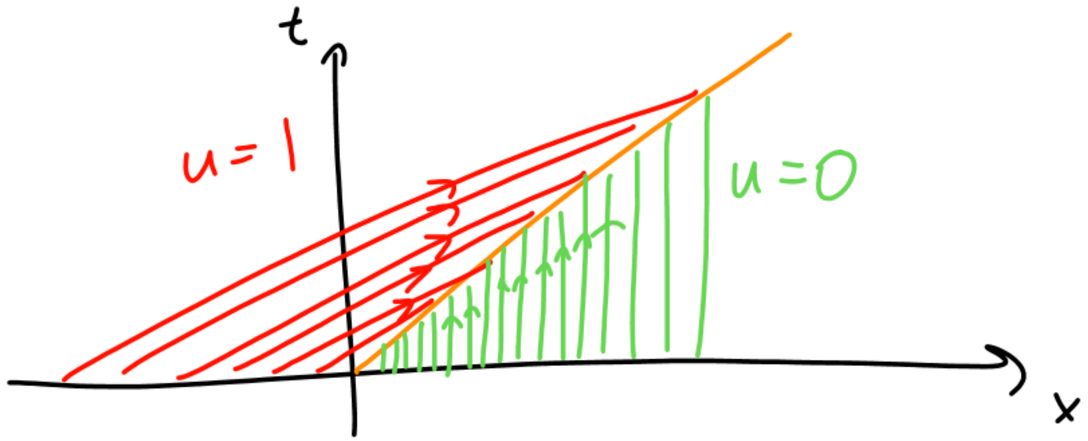
$$c'(t) = \frac{\frac{1}{2}u_\ell^2 - \frac{1}{2}u_r^2}{u_\ell - u_r}$$

$$= \frac{1}{2}\frac{(u_\ell - u_r)(u_\ell + u_r)}{u_\ell - u_r}$$

$$= \frac{u_\ell + u_r}{2}$$

If $u_\ell = 1$, $u_r = 0$,

curve $c(t) = \frac{t}{2}$

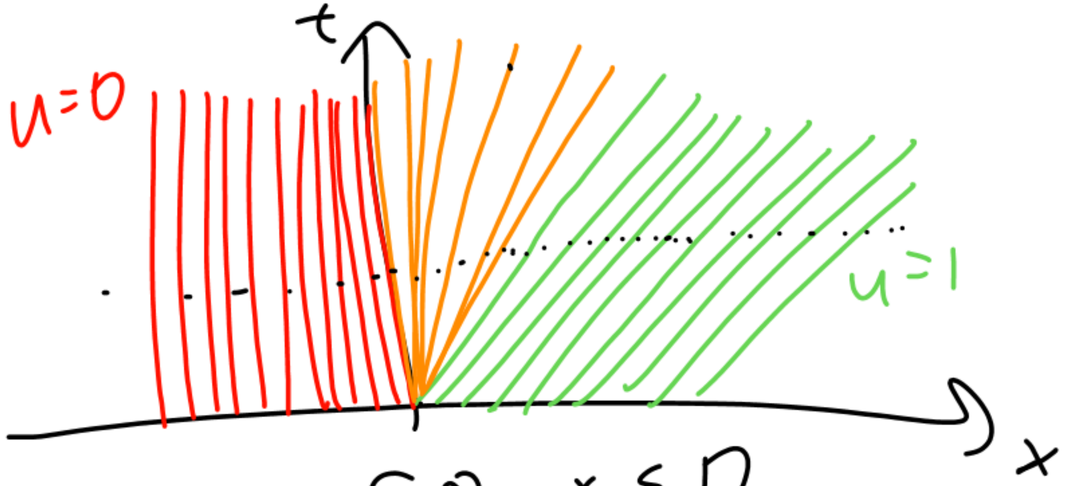


Physical shock: characteristics start at $t=0$, end at the shock.

Opposite case:

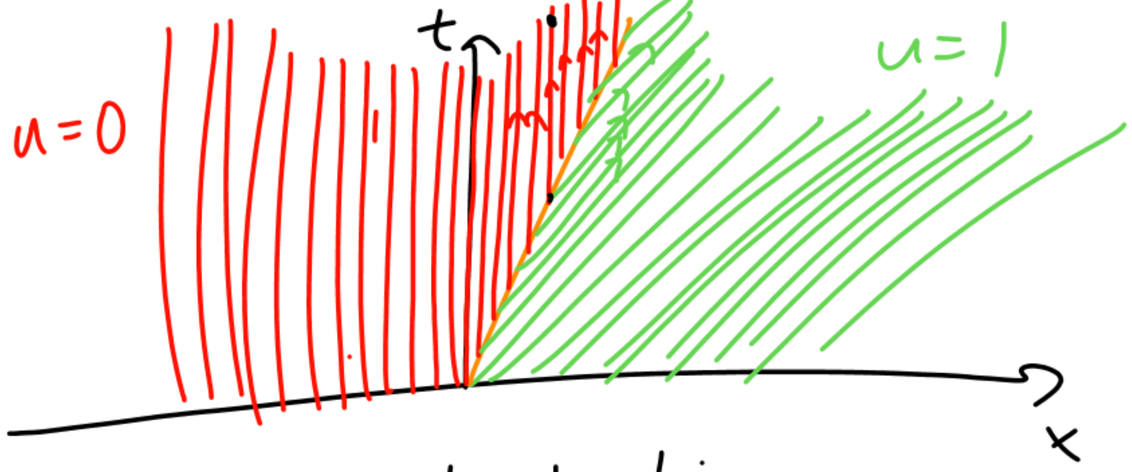
$$u(x,0) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

Rarefaction wave solution:



$$u(x,t) = \begin{cases} 0, & x < 0 \\ \frac{x}{t}, & 0 < x < t \\ 1, & x > t \end{cases}$$

An alternative solution:



Non-physical shock: characteristics originate at the shock.

Nonphysical shocks can happen when $F'(u_l) < F'(u_r)$ $(u_l < u_r)$

$\rightarrow$ In this case, use rarefaction wave.

Shocks are physical if $F'(u_\ell) > \sigma > F'(u_r)$, where σ = slope of shock

This is called the <u>entropy condition</u>

Weak Solutions

Benefits of weak solutions:

1. Definition makes sense when "pointwise" form of PDE doesn't, e.g. when some partial derivatives of u aren't defined.
2. They can carry extra info (e.g. jump conditions)
3. Sometimes, it's easier to prove existence of a weak solution.

Point ③ motivates the question:
when are weak solutions actually strong/classical/pointwise solutions?

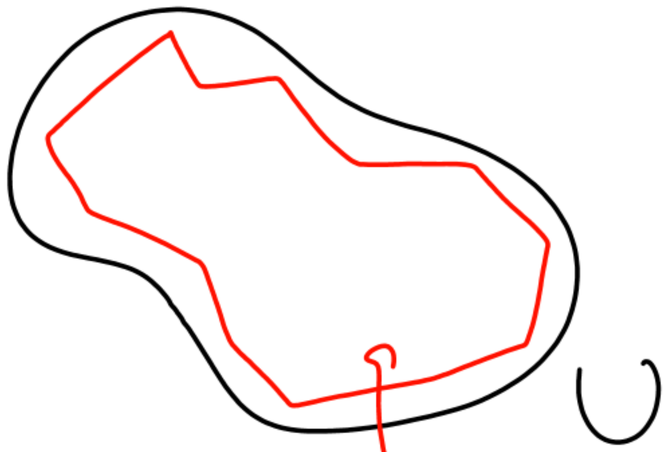
Let's look at an example:

Recall: u is <u>harmonic</u> if

$$\Delta u = 0.$$

<u>Def</u> u is <u>weakly harmonic</u> in $U \subset \mathbb{R}^n$ if, for all $\phi \in C^\infty(U)$ with compact support in U,

$$\int_U u \Delta \phi \, dx = 0$$



support(ϕ)

$\Rightarrow \phi = 0$ and $D\phi = 0$ on ∂U

<u>Thm</u> If u is continuous and weakly harmonic in U, then u is harmonic in U.

<u>Pf</u> Let u_ε be the smooth approximation to u, obtained by the standard mollifier

$$u_\varepsilon = \chi_\varepsilon * u, \quad \chi_\varepsilon(x) = \frac{1}{\varepsilon^n}\chi\left(\frac{x}{\varepsilon}\right) \to \delta(x)$$

$$\longrightarrow u \quad \text{as} \quad \varepsilon \to 0$$

χ is C^∞, $\int_{B_1} \chi \, dx = 1$, $\chi = 0$ outside B_1.

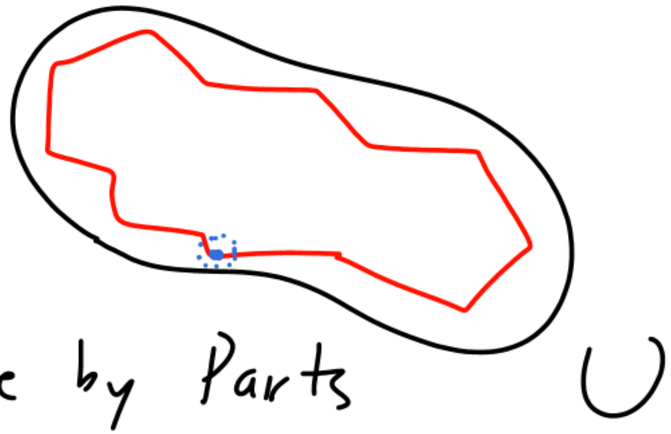
Look at

$$\int_U \Delta\phi \, u_\varepsilon \, dx,$$

with ε small enough that $B_\varepsilon(x) \subset U$ for all $x \in \text{support}(\phi)$

Integrate by Parts:

$$\int_U \Delta\phi \, u_\varepsilon \, dx = -\int_U D\phi \cdot Du_\varepsilon \, dx + \underbrace{\int_{\partial U} u_\varepsilon D\phi \cdot \nu \, dS}_{0}$$



U

Integrate by Parts again:

$$\int_U \Delta\phi \, u_\varepsilon \, dx = \int_U \phi \, \Delta u_\varepsilon \, dx - \underbrace{\int_{\partial U} \phi \, Du_\varepsilon \cdot \nu \, dS}_{0}$$

Recall: $u_\varepsilon \to u$ uniformly

$\Delta\phi$ is smooth, bounded

Dominated convergence theorem

$$\Rightarrow \int_U u_\varepsilon \Delta\phi \, dx \to \int_U u \Delta\phi \, dx = 0.$$

$\left(|u_\varepsilon \Delta\phi| \text{ uniformly bounded in } \varepsilon \text{ on } U \right)$

$$\Rightarrow \int_U \phi \Delta u_\varepsilon \, dx \to 0 \text{ as } \varepsilon \to 0.$$

$$\int \Delta\phi \, u_\varepsilon \, dx = \int_U \Delta\phi(x) \int_{\mathbb{R}^n} u(y) \, \eta_\varepsilon(x-y) \, dy \, dx$$

$$= \int_{\mathbb{R}^n} \left[\int_U \eta_\varepsilon(x-y) \, \Delta\phi(x) \, dx \right] u(y) \, dy$$

$$= \int_{\mathbb{R}^n} (\eta_\varepsilon * \Delta\phi)(y) \, u(y) \, dy \longrightarrow \int_{\mathbb{R}^n} \Delta\phi(y) \, u(y) \, dy$$

$= 0$ since u weakly harmonic.